IE 643 — DEEP LEARNING: THEORY AND PRACTICE · COURSE PROJECT

Predicting Geometric Edit Parameters from Natural Language

Preparatory Assessment · Indian Institute of Technology Bombay

Team DyoOptix

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OVERVIEW

Outline

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03	**Codebases Surveyed**	Tools evaluated for the implementation
04	**Dataset Design**	Why we generate, the grammar, magnitude distributions, splits
05	**Planned Approach**	Parameterisation, architecture, and the gated multi-head model
06	**Evaluation & Experiments**	Metrics, ablation, failure analysis and qualitative results
07	**Roadmap & Milestones**	Dated plan to the intensive assessment

BLOCK 01

Problem Statement

Given a natural-language edit instruction, predict the numeric geometric transform it implies.

"nudge the shape slightly to the left"	→	`t ≈ (−0.04, 0)`
"rotate it by 45 degrees clockwise"	→	`θ = −45°`
"make it significantly bigger"	→	`log s ≈ +0.62`

We target a 2D similarity transform — translation, rotation and uniform scale — represented as a 3×3 homogeneous matrix and applied with a deterministic affine warp.

Scope. The model predicts parameters from the instruction text alone. Those parameters are then applied to images and segmentation masks, and exposed through the interface — all delivered by the intensive assessment.

BLOCK 01

Research Questions

Every experiment in the plan exists to answer one of these three questions.

RQ1

Representation

Do frozen text representations preserve the fine-grained geometric information required to predict continuous transformation parameters?

Tested by: ablation over hand features · CLIP · MiniLM (core) and TF-IDF · fine-tuned MiniLM (additional), plus the embedding-distance diagnostic

RQ2

Language generalisation

Can the model generalise to unseen linguistic expressions while preserving magnitude?

Tested by: held-out templates · held-out adverbs · LLM paraphrases · human-written instructions

RQ3

Linguistic vagueness

What accuracy is achievable when the language specifies only a range of valid transformations rather than a single value?

Tested by: explicit vs vague magnitude subsets, measured against the noise floor

BLOCK 01

Why Parameters, Not Pixels

Generative editing

e.g. InstructPix2Pix, MagicBrush

- Generates the edited pixels directly
- No explicit geometric parameter interface
- No parameterisation whose inverse can undo the edit
- The geometric intent is not directly inspectable
- Object appearance can drift between before and after

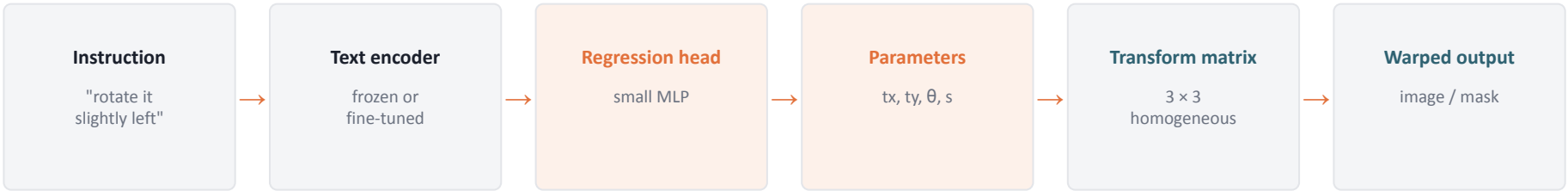
Parametric editing (ours)

text → transform parameters → deterministic warp

- Interpretable: five continuous values, directly readable
- Explicit: a requested 30° is represented as 30°
- Reversible at the transform level: M^{-1} undoes the geometry
- Auditable and composable across successive edits
- No semantic generation — deterministic geometric resampling

Wider framing. This is language grounded into structured control parameters — the same pattern as an LLM emitting a typed function call rather than prose. Geometry is a testbed where the ground truth is exact and the evaluation is unambiguous.

Formal Task Definition



Target vector

$$y = [\, t_x,\ t_y,\ \cos\theta,\ \sin\theta,\ \log s \,] \qquad \text{presence mask } p = [\, p_{trans},\ p_{rot},\ p_{scale} \,]$$

The presence mask records which transform types the instruction actually mentions, so the regression loss can ignore the rest.

Affine Geometry & the Similarity Subgroup

Why 2D needs a 3×3 matrix

A 2×2 matrix always maps the origin to the origin, so translation cannot be expressed. Lifting the point (x, y) to (x, y, 1) lets the third column inject a constant offset — translation becomes a multiplication like any other transform.

Payoff: composition

Several transforms compose by matrix product rather than by tracking a matrix plus a separate offset:

$$M = T(c) \cdot T(t) \cdot R(\theta) \cdot S(s) \cdot T(-c)$$

Centring at the object centroid c, so rotation and scale act about the object rather than the image origin.

Implementation trap: the image y-axis points down

Substituting $y \to -y$ into the standard counter-clockwise rotation gives $R = [[\cos\theta, \sin\theta], [-\sin\theta, \cos\theta]]$ — the transpose of the textbook form. "Rotate counter-clockwise" therefore carries the opposite sign to the one written from memory.

What we take from it Our output is the similarity subgroup — translation, rotation, uniform scale: 4 of affine's 6 degrees of freedom. Shear and anisotropic scale are outside the scope of the allotted task.

Spatial Transformer Networks

Jaderberg, Simonyan, Zisserman & Kavukcuoglu — NeurIPS 2015 · arXiv:1506.02025

1. **Localisation network**

   Predicts the parameters of an affine transformation from the input feature map

2. **Grid generator**

   Builds the sampling grid implied by those parameters

3. **Differentiable sampler**

   Interpolates the input at the grid points, so gradients flow through the warp

The module is inserted inside a CNN and trained end-to-end from class labels alone — no supervision on the transform itself.

What we take from it Closest architectural ancestor: a network that emits affine parameters. It differs in two ways we exploit — it conditions on an image, not text, and it has no direct parameter supervision, whereas ours has exact targets.

CLIP — the Text Encoder

Radford et al. — ICML 2021 · arXiv:2103.00020

How the text tower works

- Transformer over BPE tokens, 77-token context
- The whole sentence is pooled into one vector at the end-of-text token
- Trained contrastively against an image tower on 400M image–caption pairs
- Output: a single 512-d embedding (ViT-B/32)

Hypothesis: where CLIP may struggle

The contrastive objective optimises for semantic similarity. "Rotate by 30 degrees" and "rotate by 40 degrees" are near-paraphrases, so the training pressure pushes their embeddings together.

But the required outputs differ by 10°.
The representation geometry may therefore work against the regression.

Magnitude may be only weakly represented in the pooled embedding — this is our hypothesis, not an established result.

What we take from it Our primary frozen baseline, and the source of RQ1. We plan to test the hypothesis directly rather than assume it.

Sentence Embeddings

Reimers & Gurevych, Sentence-BERT — EMNLP 2019 · arXiv:1908.10084

What it changes

- Siamese / triplet fine-tuning of BERT for sentence-level similarity
- Mean pooling over token outputs rather than a single sentinel token
- SBERT paper: NLI and semantic-similarity data (our model's data: next slide)
- all-MiniLM-L6-v2: 384-d, 6 layers, fast on CPU

Why include it alongside CLIP

- Text–text contrastive training, vs CLIP's image–text training
- Mean pooling may retain more token-level detail than EOT pooling
- Text-only supervision, so no visual-grounding confound
- If both fail in the same way, the limitation is general, not CLIP-specific

What we take from it A second frozen baseline. If both encoders fail in the same way, the limitation is general to similarity-trained embeddings rather than specific to CLIP.

Pre-trained Models & Their Training Data

For each pre-trained encoder: what it was originally trained on, and how we apply it to our data.

CLIP ViT-B/32 — text encoder

OpenCLIP weights · laion2B-s34B-b79K

Originally trained on	LAION-2B: ~2 billion English image–text pairs crawled from the web. (The CLIP paper's model used 400M pairs; we use OpenCLIP's open reproduction.)
Objective	Contrastive, image ↔ text
How we use it	Frozen: each instruction → one 512-d vector → regression head

all-MiniLM-L6-v2

sentence-transformers

Originally trained on	MiniLM-L6 base model, fine-tuned on over 1 billion sentence pairs from multiple concatenated datasets
Objective	Contrastive, text ↔ text: find each sentence's true partner within the batch
How we use it	Frozen: each instruction → one 384-d vector → regression head; a fine-tuned variant is also compared

Our data & its relevance. Both encoders learned general English from large-scale contrastive pairs, but neither was ever trained to map magnitude words to numbers. Our ≈20k generated instruction–parameter pairs (Dataset Design) supply exactly that mapping to the regression head — and test whether the frozen embeddings preserve it (RQ1).

Do These Encoders Keep Structure?

Yuksekgonul, Bianchi, Kalluri, Jurafsky & Zou — ICLR 2023 · arXiv:2210.01936

The ARO benchmark — Attribution, Relation, Order — probes compositional understanding in vision-language models across more than 50,000 test cases.

Relation

Poor relational understanding — "a dog behind a tree" vs "a tree behind a dog"

Attribution

Models blunder when linking objects to their attributes

Order

Severe lack of word-order sensitivity

Our extension. ARO tests whether these encoders preserve relations, attributes and order. It does not test whether they preserve ***magnitude***. That is precisely the property our task depends on, and the gap our first experiment targets.

What we take from it Prior work reports weaknesses in compositional structure for some vision-language models. We test whether fine-grained geometric magnitude is similarly weakly represented (RQ1).

BLOCK 03

Codebases & Tools Surveyed

TOOL	ROLE	STATUS	REASONING
OpenCLIP	Frozen CLIP text tower	**Adopted**	Cleaner access to the text encoder alone than the original repo; pinned ViT-B/32
sentence-transformers	Frozen sentence embeddings	**Adopted**	One-line API for all-MiniLM-L6-v2; CPU-friendly
OpenCV	Affine warping	**Adopted**	getRotationMatrix2D matches our convention; will unit-test our matrix builder
kornia	Differentiable warps in PyTorch	**Not required**	Only needed for an end-to-end pixel loss; OpenCV covers our warping
SAM	Object masks for real images	**Not required**	COCO ships segmentation masks, so no extra segmentation model is needed
Gradio	Demo interface	**Milestone 7**	Required deliverable; simplest route to an upload-and-edit demo

Progress so far: environment built and tested (CLIP → 512-d, MiniLM → 384-d, OpenCV warp working); problem, dataset and evaluation designed. Not yet: data generation, training or results.

Why We Generate the Dataset

We did not identify a dataset that fits this task

- Instruction-editing corpora (e.g. InstructPix2Pix, MagicBrush) provide (image, instruction, edited image) triples
- The numeric transform was never recorded — these systems generate pixels directly and never compute one
- Recovering parameters from before/after pairs is a harder problem than the one we are solving

Our approach: invert the labelling order

Sample the parameter first, then construct the sentence around it.

Annotation is normally the expensive step. Generating the number before the text makes labels exact and free — there is no annotator disagreement, no labelling budget and no measurement noise beyond what we deliberately introduce.

Existing corpora are still used — as a reference for realistic phrasing, so our templates do not drift into unnatural English. Real images and masks (COCO) supply the pictures to which the predicted transforms are applied.

Generation Grammar

VERB

determines transform type

move · shift · nudge · slide · drag
rotate · turn · spin · tilt
scale · resize · enlarge · shrink

DIRECTION

determines sign and axis

left · right · up · down
upper-left · toward the bottom-right
clockwise · anticlockwise

MAGNITUDE

determines the number

vague: slightly · moderately ·
considerably · significantly · drastically
explicit: by 30 px · by 45° · by 2×

Worked example

"nudge the shape slightly to the left"

→ `{ tx : −0.043, ty : 0.0 }, presence [1, 0, 0]`

Scale

≈ 20,000

instruction–parameter pairs

Deliberate hard cases: compositional ("move it right and shrink it") · partial ("just rotate it, don't move it") · negation ("don't rotate it") · paraphrase ("turn it halfway around" = 180°)

Excluded by design: context-dependent instructions ("centre it"), whose target depends on where the object currently is and so cannot be determined from text alone, and reflections ("flip it"), which lie outside the similarity subgroup (det M < 0).

Magnitude Is a Distribution, Not a Number

ADVERB	TRANSLATION (frac. width)	ROTATION	|log s|
slightly / a bit	U(0.02, 0.06)	U(3°, 10°)	U(0.05, 0.15)
moderately	U(0.10, 0.18)	U(15°, 30°)	U(0.20, 0.40)
significantly	U(0.22, 0.32)	U(40°, 70°)	U(0.50, 0.80)
drastically	U(0.35, 0.45)	U(75°, 110°)	U(0.90, 1.30)

"Slightly" has no correct numeric answer. Sampling rather than fixing means the labels carry irreducible noise.

Noise floor

An error level no model can beat: under MAE the optimal point predictor is the conditional median (the mean, under MSE). Absolute MAE alone is uninterpretable; we report model MAE ÷ oracle MAE.

Free ablation

Explicit magnitudes ("by 45 degrees") are near-noiseless, giving a low-noise subset to compare against the vague-adverb subset.

Novelty direction

Motivates predicting a distribution rather than a point, so the model learns that "by 45°" is confident and "slightly" is not — our planned task for the novelty assessment.

These ranges are an operational convention for dataset generation, not a claim about linguistic ground truth. What the experiments depend on is the ordering of the adverbs, not the exact endpoints.

Splits & Generalisation Tests

SPLIT	CONSTRUCTION	WHAT IT TESTS
Templated train / val	In-distribution fit	Baseline performance
Held-out templates	Unseen sentence structures	Does it generalise beyond memorised patterns?
Held-out adverbs	Train on slightly / moderately / significantly; test on a touch, considerably	Has it learned magnitude ordering, or a lookup table?
LLM-paraphrased	Templated sentences rewritten naturally	Robustness to natural phrasing
Human-written (≈100)	Written by hand by the team	True out-of-distribution check

Curation: duplicates removed, edit types and magnitudes balanced, splits made at template-family level so no near-duplicate crosses train and test. The held-out-adverb split is the discriminating one — a lookup table scores zero on it.

Parameterisation & Conventions

CHOICE	DECISION	REASONING
Translation	Fraction of image width, $[-0.5, 0.5]$	Resolution-independent; a model trained at 256 px transfers to 512 px
Rotation	Regress $(\cos\theta, \sin\theta)$, recover with atan2	Raw angle wraps at ±180°, forcing a discontinuous output (Zhou et al., CVPR 2019)
Scale	Regress $\log s$, $s \in [0.25, 4]$	$\log 2 = +0.69$, $\log 0.5 = -0.69$ — enlarging and shrinking penalised symmetrically
Isotropy	Uniform scale	Anisotropic scaling ("make it wider") lies outside the similarity transform
Centre	Object centroid	Rotating about the image origin throws the object across the frame
Composition	$M = T(c)\cdot T(t)\cdot R(\theta)\cdot S(s)\cdot T(-c)$	Order matters once scale is anisotropic; to be fixed in a written spec

Sign convention. The image y-axis points down, so the rotation matrix is the transpose of the textbook form. To be verified against cv2.getRotationMatrix2D in a unit test (milestone 1).

Planned Architecture

Representation ablation

The input side is the variable; everything downstream is held fixed.

CORE — NAMED IN THE PROJECT BRIEF

- Hand-engineered slot features + regex numerals
- Frozen CLIP text tower (512-d)
- Frozen all-MiniLM-L6-v2 (384-d)

ADDITIONAL COMPARISONS

- TF-IDF — non-neural lexical baseline
- Fine-tuned MiniLM — same model unfrozen, so only one variable changes

Gated multi-head regressor



Presence head

3 sigmoids · BCE loss

Value heads

(tx,ty) · (cos,sin) · log s

Because many instructions omit a given transform type, naïve regression is dominated by zero-valued targets: the model learns to always output zero and the error metric still looks excellent. We therefore mask the regression loss to the components the instruction actually mentions.

Central hypothesis. (RQ1) The bottleneck may be the representation rather than the head. Planned first experiment: plot embedding cosine distance against ground-truth parameter distance. A weak correlation would indicate the encoder collapses instructions requiring very different outputs.

Evaluation Metrics

The project brief allows us to define our own metrics where none exist. We plan three tiers.

Parameter-level

- MAE / RMSE per parameter, reported in interpretable units (pixels, degrees, scale ratio)
- Never blended into one number — the units are incommensurable
- Edit-type F1 from the presence head
- Direction sign accuracy (left vs right) reported separately from magnitude error

Geometric / end-task

- Corner displacement: transform the object's bounding-box corners twice — once with the predicted matrix, once with the true one — and average the distance between the two results. Dividing by the object's diagonal makes the score independent of object size. Follows the principle used in ADD-style pose evaluation
- Mask IoU: warp the same mask with the predicted matrix and with the true matrix, then measure intersection over union between the two warped masks. Bounded in [0,1] and visually intuitive

Proposed (ours)

- Ordinal consistency (RQ2): Spearman correlation between adverb intensity rank and predicted magnitude — tests whether the scale of language was learned
- Normalised error vs noise floor (RQ3): the model's error divided by the best error achievable given the ambiguity, where that best predictor is the conditional median. A value near 1.0 indicates the model is approaching the noise floor

Rejected: the Frobenius norm of the matrix difference — it mixes translation, rotation and scale units into one uninterpretable quantity.

Ablation, Failure Analysis & Qualitative Results

Ablation & breakdown

Vary the text representation, hold the regression head fixed, and report every result broken down:

- By edit type — translation / rotation / scale / composite
- By magnitude — explicit / small / medium / large
- By split — seen vs unseen instructions

Failure cases we will look for

- Sign flips — moves right when told left
- Magnitude collapse — every prediction near the average
- Composite instructions — only the first edit applied
- Negation ignored — "don't rotate it" still rotates
- Explicit numbers ignored — "by 45°" treated as a generic rotate

Qualitative results

- Apply predicted edits to sample images and masks — synthetic shapes first, then COCO
- Show original / ground truth / prediction side by side
- Include failure cases, not just successes

Interface (Gradio)

- Upload an image and mask, type an instruction
- Shows the predicted parameters, the edited image, and IoU against ground truth where available

BLOCK 06

Roadmap & Milestones

#	MILESTONE	TARGET DATES	OWNER	RISK
1	Transform specification, matrix builder, unit tests	22 – 24 Sep	K	Low
2	Dataset generator, splits, ≈20k pairs	24 – 28 Sep	K	Low
3	Noise-floor computation; metric module	28 Sep – 2 Oct	A	Low
4	Representation ablation, diagnostic plot, failure analysis	2 – 8 Oct	A	Medium
5	Gated multi-head model; training loop	8 – 15 Oct	K	Medium
6	Real images (COCO masks), renderer, qualitative grid	15 – 22 Oct	A	Medium
7	Gradio interface	22 – 27 Oct	K	Low
8	Report, code documentation, walkthrough video	27 Oct – 1 Nov	Both	Low
9	Novelty assessment: uncertainty-aware prediction	after 1 Nov	Both	High

K = Kaushik, A = Akash

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